Type II_∞ for the gravitational algebra of a comoving observer in radiation-dominated FRW via conformal pullback*

Kevin Remondière
Independent researcher
kevin.remondiere@gmail.com
ORCID: 0009-0008-2443-7166

2026-05-02

Abstract

We observe that, for the conformally coupled massless scalar field on a radiation-dominated FRW spacetime ($a(\eta) = a_0\eta$), the local algebra $\mathcal{A}(\mathcal{O})_{\mathrm{FRW}}$ associated to a causal diamond of a comoving observer with $\eta_i > 0$ (away from the Big Bang) is *unitarily* equivalent, as a W^*-algebra equipped with its vacuum state, to the corresponding Minkowski local algebra $\mathcal{A}(M_{\mathcal{O}})_{\mathrm{M}}$ on the conformally identified diamond. The unitary is the second-quantised version of the standard field rescaling $\varphi \mapsto a(\eta)\,\varphi$; we verify symbolically (sympy) the off-shell conformal-covariance identity for the wave operator and the cancellation of $\partial_\eta a$ derivative terms in the symplectic form on a Cauchy slice. By Tomita–Takesaki uniqueness, the modular automorphism group of $(\mathcal{A}(\mathcal{O})_{\mathrm{FRW}}, \Omega_{\mathrm{FRW}})$ is therefore unitarily intertwined with the Hislop–Longo modular flow of the Minkowski diamond in the vacuum, hence has Connes spectrum $\Gamma(\sigma) = \mathbb{R}$. Consequently the crossed product $\mathcal{A}(\mathcal{O})_{\mathrm{FRW}} \rtimes_\sigma \mathbb{R}$ is type II_∞. This sidesteps the stationarity hypothesis (Killing vector field) of Chandrasekaran–Longo–Penington–Witten 2022 and gives the first explicit non-stationary cosmological example of the CLPW type II_∞ classification. The argument extends verbatim to matter-dominated FRW ($a(\eta) \propto \eta^2$, Theorem 6) and to slow-roll inflation modelled by exact de Sitter ($a(\eta) = -1/(H\eta)$, Theorem 7), provided the diamond is bounded away from $\eta = 0$ in each case so that $a(\eta)$ is smooth and strictly positive on the closure. The argument is strictly limited to the conformally coupled massless scalar on a conformally flat background: generic FRW backgrounds (with non-conformal matter content), fields with $\xi \neq 1/6$ or $m \neq 0$, and wedge-type regions extending to $\eta = 0$ are explicitly outside its scope.

1 Introduction

The crossed-product construction of Witten [10] and Chandrasekaran–Longo–Penington–Witten (CLPW) [3] produces, from the type III_1 local algebra of a free quantum field theory and a suitable modular automorphism group, a type II von Neumann algebra of *gravitationally dressed* observables. The key ingredient is a one-parameter group of automorphisms with full Connes spectrum $\Gamma(\sigma) = \mathbb{R}$, which under crossed-product yields a type II_∞ factor (semifinite, with a well-defined trace up to scaling). For static patches of de Sitter space CLPW [3] obtain instead a type II_1 factor by including the constraint that the modular Hamiltonian of the static patch is bounded below. Faulkner–Speranza [5] extend the crossed-product framework to general Killing horizons and prove a non-perturbative generalised second law in this setting.

*Standalone technical note. The result was first recorded in commit `aa9a3ee` of https://github.com/AIdevsmartdata/crossed-cosmos (2026-05-02).

In all of these constructions, the modular automorphism is identified with the flow of a Killing vector field of the background spacetime: the boost in Minkowski (Bisognano–Wichmann) [1,2], the de Sitter boost in the static patch [3], the Killing horizon generator in [5]. In other words, all known cases of the CLPW type-II classification rely, at the present level of the literature, on the existence of a stationary Killing vector field whose flow geometrically implements the modular automorphism.

The purpose of this note is to record one example where this stationarity hypothesis can be removed. We consider a radiation-dominated FRW spacetime, which has *no* timelike Killing vector field along a generic comoving worldline, but *is* conformally flat. For the conformally coupled massless scalar field, the conformal isometry to Minkowski space lifts to a unitarily implemented W*-algebraic isomorphism of local algebras, intertwining the FRW vacuum (conformal vacuum) and the Minkowski vacuum. This reduces the FRW modular flow to the Minkowski Hislop–Longo [7] modular flow on the conformally identified diamond. Tomita–Takesaki uniqueness then yields the type II_∞ classification of the crossed product, with no Killing vector required.

Statement (informal, see Theorem 5). *Let φ be the conformally coupled massless scalar field on radiation-dominated FRW with $a(\eta) = a_0\eta$ on $\eta > 0$, in the conformal vacuum. Let $\mathcal{O} = D_{\mathcal{O}}$ be a causal diamond of a comoving observer with $0 < \eta_i < \eta_f < \infty$, and $\mathcal{A}(\mathcal{O})$ the associated local von Neumann algebra. Then $\mathcal{A}(\mathcal{O}) \rtimes_\sigma \mathbb{R}$ is a type II_∞ factor.*

Scope and limitations. The argument uses conformal flatness of the FRW background and conformal invariance of the field equation $(\Box_g - R/6)\varphi = 0$ in $d = 4$. It applies as written to any conformally flat FRW background with a strictly positive scale factor on the closed diamond, including matter-dominated FRW with $a(\eta) \propto \eta^2$ and slow-roll inflation modelled by exact de Sitter $a(\eta) = -1/(H\eta)$. We restrict throughout to *causal diamonds* $D_{\mathcal{O}}$ with $\eta_i > 0$ (resp. $\eta_f < 0$ for de Sitter): the wedge case is excluded because a Rindler-type wedge in conformal coordinates extends to $\eta \to 0$ and unavoidably crosses the singularity (or, in the de Sitter case, the conformal boundary). The argument does *not* apply to generic FRW backgrounds beyond conformal flatness, nor to massive or non-conformal scalar fields, nor to interacting fields. We discuss these limitations and the corresponding open problems in Section 5.

2 Preliminaries

2.1 Conformal flatness of radiation-dominated FRW

A spatially flat FRW spacetime in conformal time η has metric

$$\mathrm{d}s^2_{\mathrm{FRW}} = a(\eta)^2\big(-\mathrm{d}\eta^2 + \mathrm{d}\mathbf{x}^2\big), \qquad \mathbf{x} \in \mathbb{R}^3. \tag{1}$$

For radiation-dominated matter content ($a^4\rho = \text{const}$, $p = \rho/3$), the Friedmann equations integrate to

$$a(\eta) = a_0\,\eta, \qquad \eta \in (0, \infty). \tag{2}$$

The Big Bang is at $\eta = 0$. The metric (1) is, by inspection, $a(\eta)^2$ times the Minkowski metric $\eta_{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1)$. This is the standard fact that any spatially flat FRW spacetime is conformally flat in conformal coordinates [9, §5.1, §5.3].

For matter-dominated FRW ($a^3\rho = \text{const}$, $p = 0$), the same coordinate change gives $a(\eta) \propto \eta^2$, again conformally flat. Generic mixed FRW backgrounds with non-conformal matter content (e.g. a cosmological constant + dust) are likewise conformally flat as Lorentzian manifolds, since spatial-flatness plus FRW symmetry forces the Weyl tensor to vanish; the conformal factor $a(\eta)$ is then determined by integrating the Friedmann equation.

2.2 Conformally coupled massless scalar

The conformally coupled massless scalar in $d = 4$ has classical action (Wald sign convention [9, §4.6])

$$S[\varphi] = -\tfrac{1}{2} \int \mathrm{d}^4x \, \sqrt{-g} \left[g^{\mu\nu} \partial_\mu \varphi \, \partial_\nu \varphi - \tfrac{1}{6} \, R \, \varphi^2 \right], \tag{3}$$

with $\xi = 1/6$ the conformal coupling. Its equation of motion is $(\Box_g - R/6)\varphi = 0$. Under a Weyl rescaling $g_{\mu\nu} = \Omega^2 \tilde{g}_{\mu\nu}$ with $\Omega = a(\eta)$ and $\tilde{g}_{\mu\nu}$ the Minkowski metric, the conformally rescaled field

$$\tilde{\varphi} \; := \; a(\eta) \, \varphi \tag{4}$$

satisfies the flat wave equation $\Box_{\tilde{g}} \, \tilde{\varphi} = 0$ on Minkowski space, as a consequence of the off-shell conformal-covariance identity

$$\Box_{\tilde{g}} \big(a\varphi \big) \; = \; a^3 \, \big(\Box_g - R/6 \big) \varphi \tag{5}$$

which is the standard conformal covariance of the $d = 4$ conformally coupled scalar [9, §4.6]. We re-verify (5) symbolically (sympy) in Section 3.1.

2.3 Local algebras and Bunch–Davies vacuum

We work in the algebraic framework. To each open relatively compact region $\mathcal{O}$ of the FRW spacetime with causally complete closure we associate the (smeared) field algebra

$$\mathcal{A}(\mathcal{O})_{\mathrm{FRW}} \; := \; \pi_\Omega \big(\{ \varphi(f) : f \in C_c^\infty(\mathcal{O}) \} \big)'', \tag{6}$$

the double commutant in the GNS representation π_Ω of the Bunch–Davies (= conformal) vacuum state Ω_{FRW}. Standard Haag–Kastler axioms hold in this representation; existence of the Bunch–Davies vacuum for the conformally coupled massless scalar in conformally flat FRW is classical [9, §4.6].

The conformal vacuum is, by definition, the state whose two-point function pulls back to the Minkowski Wightman function under the rescaling (4): $\langle \varphi(x)\varphi(y) \rangle_{\Omega_{\mathrm{FRW}}} = a(\eta_x)^{-1} a(\eta_y)^{-1} \langle \tilde{\varphi}(x)\tilde{\varphi}(y) \rangle_{\Omega_{\mathrm{M}}}$. This identifies Ω_{FRW} with the Minkowski vacuum after the field redefinition (4). Equivalently, the GNS Hilbert spaces are canonically identified by a unitary $U : \mathcal{H}_{\mathrm{FRW}} \to \mathcal{H}_{\mathrm{M}}$ implementing $\varphi(f) \mapsto \tilde{\varphi}(a \cdot f)$, with $U\Omega_{\mathrm{FRW}} = \Omega_{\mathrm{M}}$ (see §3.2).

2.4 Diamond algebras (Hislop–Longo) and Bisognano–Wichmann background

For a doubly-bounded causal diamond $D \subset \mathbb{R}^{1,3}$ (the causal completion of an open ball on a Cauchy slice), the modular structure is given by the Hislop–Longo theorem [7]: for the free massless scalar field, the modular automorphism group of $(\mathcal{A}(D), \Omega_{\mathrm{M}})$ is implemented by a one-parameter subgroup of conformal transformations of Minkowski space (a Möbius flow on the diamond fixing its tips), and $\mathcal{A}(D)$ is a type III_1 factor with full Connes spectrum $\Gamma(\sigma) = \mathbb{R}$. This is the regime in which the present note operates.

For background and contrast we recall Bisognano–Wichmann [1, 2]. In Minkowski space, fix a Rindler wedge $W_R = \{x \in \mathbb{R}^{1,3} : x^1 > |x^0|\}$. The Tomita–Takesaki modular operator of the pair $(\mathcal{A}(W_R)_{\mathrm{M}}, \Omega_{\mathrm{M}})$ has modular automorphism group given by the boost flow,

$$\sigma_s^{\mathrm{BW}}(\varphi(x)) = \varphi(\Lambda_{2\pi s} \, x), \tag{7}$$

where Λ_t is the Lorentz boost in the (x^0, x^1)-plane with rapidity t, and $\Delta_{\mathrm{BW}} = \exp(-2\pi K_{\mathrm{BW}})$ with K_{BW} the boost generator restricted to W_R.

II_∞ vs II_1 via the modular Hamiltonian spectrum. The CLPW [3] type-II classification distinguishes type II_1 from type II_∞ by whether the modular Hamiltonian $K = -\log\Delta$ is bounded below (yielding II_1, as in the de Sitter static patch) or unbounded both ways (yielding II_∞). For the Minkowski Rindler wedge in the vacuum, the modular Hamiltonian is the boost generator K_{BW}, whose self-adjoint spectrum is all of $\mathbb{R}$ (the boost generator on Minkowski space has continuous unbounded spectrum on both ends). Hence the Connes spectrum $\Gamma(\sigma^{\mathrm{BW}}) = \mathbb{R}$ and the Connes–Takesaki crossed product $\mathcal{A}(W_R) \rtimes_{\sigma^{\mathrm{BW}}} \mathbb{R}$ is a type II_∞ factor. The same holds for the Hislop–Longo modular flow on a Minkowski diamond, since the Möbius-flow generator (a conformal Killing field) is conjugate to a boost generator in the $\mathrm{SO}(2,4)$ conformal group of $\mathbb{R}^{1,3}$ and shares the same spectral type. Throughout the present note, we work in the II_∞ regime; II_1 would only arise if the modular Hamiltonian on the FRW diamond were bounded below, which is not the case once it is conjugated to the Möbius generator.

2.5 The crossed product

Given a von Neumann algebra $\mathcal{M}$ and a one-parameter group of automorphisms $\sigma : \mathbb{R} \to \mathrm{Aut}(\mathcal{M})$, the crossed product $\mathcal{M} \rtimes_\sigma \mathbb{R}$ is the von Neumann algebra on $L^2(\mathbb{R}, \mathcal{H})$ generated by $\pi(m)\xi(s) = \sigma_{-s}(m)\xi(s)$ and translation $u_t\xi(s) = \xi(s-t)$. When $\mathcal{M}$ is type III_1 and σ is a modular automorphism with full Connes spectrum, $\mathcal{M} \rtimes_\sigma \mathbb{R}$ is a type II_∞ factor (Connes–Takesaki). This is the construction underlying CLPW [3, 10].

3 Main theorem

3.1 The symplectic form on FRW (sympy-verified)

We verify off-shell and explicitly the conformal-covariance identity (5) and its consequence that the field rescaling $\varphi = \tilde{\varphi}/a$ matches the FRW symplectic form to the Minkowski symplectic form, with no leftover $\partial_\eta a$ term. Both calculations are double-checked symbolically in sympy 1.12; the scripts (`symplectic_check.py`, `conformal_box_check.py`) are bundled with the source.

(a) Conformal covariance of the wave operator. For the metric $g_{\mu\nu} = a(\eta)^2\,\eta_{\mu\nu}$, direct computation of the Christoffels gives

$$\Box_g\,\varphi \;=\; \frac{1}{a^4}\,\partial_\mu\big(a^4\,g^{\mu\nu}\,\partial_\nu\varphi\big) \;=\; \frac{1}{a^2}\,\Box_{\tilde{g}}\,\varphi \;-\; \frac{2\,(\partial_\eta a)}{a^3}\,\partial_\eta\varphi, \tag{8}$$

and the Ricci scalar of g in conformal coordinates is $R = 6\,a^{-3}\,(\partial_\eta^2 a)$ ([9, §5.1]; we re-derived it from the Christoffels in our sympy script). Combining (8) with $\Box_{\tilde{g}}(a\,\varphi) = a\,\Box_{\tilde{g}}\,\varphi - 2(\partial_\eta a)\partial_\eta\varphi - (\partial_\eta^2 a)\,\varphi$ gives, off-shell and exactly,

$$\Box_{\tilde{g}}\big(a\,\varphi\big) \;=\; a^3\,\Big(\Box_g\,\varphi \;-\; \tfrac{1}{6}\,R\,\varphi\Big), \tag{9}$$

which is (5). *Sympy verification.* The script `conformal_box_check.py` computes both sides of (9) in $(\eta, \mathbf{x})$ components for arbitrary smooth $a(\eta)$ and arbitrary $\varphi(\eta, \mathbf{x})$, and reports a difference of zero after `simplify`. The sign of the curvature term in the action and EOM (3) matches Wald's convention [9, §4.6].

(b) Symplectic form. On a constant-η Cauchy slice $\Sigma_\eta \subset D_{\mathcal{O}}$, the canonical symplectic form for the conformally coupled scalar (Klein–Gordon current $j_\mu = \varphi_1\,\partial_\mu\varphi_2 - \varphi_2\,\partial_\mu\varphi_1$, contracted with the unit normal and integrated against the induced volume) reads

$$\sigma_{\mathrm{FRW}}(\varphi_1, \varphi_2) \;=\; \int_{\Sigma_\eta} \big(\varphi_1\,\partial_\eta\varphi_2 - \varphi_2\,\partial_\eta\varphi_1\big)\,a^2\,\mathrm{d}^3\mathbf{x}, \tag{10}$$

where the weight a^2 comes from $\sqrt{h}\, n^\eta = a^3 \cdot a^{-1} = a^2$, with $\sqrt{h} = a^3$ the induced 3-volume and $n^\mu = (1/a, \mathbf{0})$ the future-directed unit timelike normal in g_{FRW}. (The curvature term $-(R/6)\varphi^2$ in the action is symmetric in any product $\varphi_1\varphi_2$ and contributes nothing to the antisymmetrised current; the symplectic form for a conformally coupled scalar coincides with the standard Klein–Gordon form on a Cauchy slice.) Substituting $\varphi_i = \tilde{\varphi}_i/a$:

$$\begin{aligned}
\sigma_{\mathrm{FRW}}(\varphi_1, \varphi_2) &= \int_{\Sigma_\eta} a^2 \Big[\tfrac{\tilde{\varphi}_1}{a} \partial_\eta\big(\tfrac{\tilde{\varphi}_2}{a}\big) - \tfrac{\tilde{\varphi}_2}{a} \partial_\eta\big(\tfrac{\tilde{\varphi}_1}{a}\big) \Big] \mathrm{d}^3\mathbf{x} \\
&= \int_{\Sigma_\eta} \Big[\tilde{\varphi}_1 \Big(\tfrac{\partial_\eta \tilde{\varphi}_2}{a} - \tfrac{(\partial_\eta a)\tilde{\varphi}_2}{a^2} \Big) \cdot a - \tilde{\varphi}_2 \Big(\tfrac{\partial_\eta \tilde{\varphi}_1}{a} - \tfrac{(\partial_\eta a)\tilde{\varphi}_1}{a^2} \Big) \cdot a \Big] \mathrm{d}^3\mathbf{x} \\
&= \int_{\Sigma_\eta} \big(\tilde{\varphi}_1 \partial_\eta \tilde{\varphi}_2 - \tilde{\varphi}_2 \partial_\eta \tilde{\varphi}_1 \big) \mathrm{d}^3\mathbf{x} \; - \; \int_{\Sigma_\eta} \frac{\partial_\eta a}{a} \big(\tilde{\varphi}_1 \tilde{\varphi}_2 - \tilde{\varphi}_2 \tilde{\varphi}_1 \big) \, \mathrm{d}^3\mathbf{x} \\
&= \sigma_{\mathrm{M}}(\tilde{\varphi}_1, \tilde{\varphi}_2) \; - \; 0.
\end{aligned}$$

The two $(\partial_\eta a)/a$ terms cancel *by antisymmetry of the symplectic-current bracket*, since $\tilde{\varphi}_1\tilde{\varphi}_2 = \tilde{\varphi}_2\tilde{\varphi}_1$ pointwise on the Cauchy slice. *Sympy verification.* The script `symplectic_check.py` computes the integrand $a^2\big(\varphi_1\partial_\eta\varphi_2 - \varphi_2\partial_\eta\varphi_1\big) - \big(\tilde{\varphi}_1\partial_\eta\tilde{\varphi}_2 - \tilde{\varphi}_2\partial_\eta\tilde{\varphi}_1\big)$ with $\varphi_i = \tilde{\varphi}_i/a$ for arbitrary smooth $a(\eta)$ and arbitrary smooth $\tilde{\varphi}_1, \tilde{\varphi}_2$, and reports a difference of zero after `simplify`.

At the level of test-function smearings we use the convention $\varphi(f) := \int \varphi(x)\, f(x) \sqrt{-g}\, \mathrm{d}^4x$ on g_{FRW} and $\tilde{\varphi}(h) := \int \tilde{\varphi}(x)\, h(x)\, \mathrm{d}^4x$ on Minkowski. Then $\varphi(f) = \tilde{\varphi}(a^3 f)$, so the natural test-function bijection $C_c^\infty(D_\mathcal{O}) \to C_c^\infty(M_\mathcal{O})$ is $f \mapsto a^3 f$. Smoothness and strict positivity of a on $\overline{D_\mathcal{O}}$ make this a C_c^∞-bijection (its inverse is $h \mapsto a^{-3}h$, which is also C_c^∞). The symplectic compatibility on solution space (10) together with the field equation (9) ensures that the induced map on the symplectic space of test functions modulo solutions is a bijective linear $\mathbb{R}$-symplectomorphism.

3.2 The conformal unitary U

Fix a comoving observer in radiation-dominated FRW at $\mathbf{x} = 0$ and a doubly-bounded causal diamond $D_\mathcal{O}$ with conformal-time interval $[\eta_i, \eta_f]$, $0 < \eta_i < \eta_f < \infty$. In the $(\eta, \mathbf{x})$ coordinates of (1), the diamond $D_\mathcal{O}$ is the same set of coordinate points as a Minkowski diamond $M_\mathcal{O} \subset \mathbb{R}^{1,3}$. We use both descriptions: $(D_\mathcal{O}, g_{\mathrm{FRW}})$ and $(M_\mathcal{O}, \eta_{\mu\nu})$, related by $(g_{\mathrm{FRW}})_{\mu\nu} = a(\eta)^2\, \eta_{\mu\nu}$.

Proposition 1. *There exists a unitary $U : \mathcal{H}_{\mathrm{FRW}} \to \mathcal{H}_{\mathrm{M}}$ such that*

(i) $U\,\Omega_{\mathrm{FRW}} = \Omega_{\mathrm{M}}$*;*

(ii) $U\,\varphi_{\mathrm{FRW}}(f)\,U^{-1} = \tilde{\varphi}_{\mathrm{M}}(a^3 \cdot f)$ *for all* $f \in C_c^\infty(D_\mathcal{O})$*, where* $(a^3 \cdot f)(\eta, \mathbf{x}) := a(\eta)^3\, f(\eta, \mathbf{x})$*;*

(iii) $U\,\mathcal{A}(D_\mathcal{O})_{\mathrm{FRW}}\,U^{-1} = \mathcal{A}(M_\mathcal{O})_{\mathrm{M}}$ *as von Neumann subalgebras of* $B(\mathcal{H}_{\mathrm{M}})$*.*

In particular the W^*-isomorphism $\mathrm{Ad}\,U : \mathcal{A}(D_\mathcal{O})_{\mathrm{FRW}} \to \mathcal{A}(M_\mathcal{O})_{\mathrm{M}}$ *is* spatially implemented *by the unitary U, and U maps the cyclic separating vector Ω_{FRW} to the cyclic separating vector Ω_{M}. By the Tomita–Takesaki uniqueness theorem (Bratteli–Robinson II, §2.5; Takesaki II, Ch. VIII), the modular operators and modular conjugations therefore intertwine:* $\Delta_{\mathrm{FRW}} = U^{-1}\Delta_{\mathrm{M}}U$ *and* $J_{\mathrm{FRW}} = U^{-1}J_{\mathrm{M}}U$*.*

Proof sketch. By Section 3.1(b), the field rescaling $\varphi = \tilde{\varphi}/a$ together with the test-function rescaling $f \mapsto a^3 \cdot f$ matches the FRW classical symplectic form to the Minkowski one, on solutions of $D_\mathcal{O}$ and $M_\mathcal{O}$ respectively. Since $a(\eta)$ is smooth and strictly positive on the closed diamond $\overline{D_\mathcal{O}}$ (we used $\eta_i > 0$), this is a bijective linear symplectomorphism of the relevant test-function spaces.

The induced symplectic transformation on the one-particle Hilbert space is therefore a Bogoliubov transformation that preserves the (positive) one-particle two-point function, because the conformal vacuum is, by definition (Section 2.3), the state on $\mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}}$ that pulls back to the Minkowski vacuum two-point function under (4). It therefore lifts to a unitary U on the symmetric Fock spaces with $U\Omega_{\mathrm{FRW}} = \Omega_{\mathrm{M}}$, which gives (i) and the implementation property (ii). Statement (iii) follows from (ii) by taking double commutants: the image $\{\tilde{\varphi}_{\mathrm{M}}(a^3 \cdot f) : f \in C_c^\infty(D_{\mathcal{O}})\}$ generates the same von Neumann algebra as $\{\tilde{\varphi}_{\mathrm{M}}(h) : h \in C_c^\infty(M_{\mathcal{O}})\}$ since $f \mapsto a^3 f$ is a C_c^∞-bijection (its inverse being $h \mapsto a^{-3}h$, C_c^∞ by smoothness and positivity of a on $\overline{D_{\mathcal{O}}}$).

The final intertwining of modular data is then immediate: $\mathrm{Ad}\, U$ is a $*$-isomorphism between two von Neumann algebras that is spatially implemented (by U) and maps a cyclic separating vector to a cyclic separating vector. Tomita–Takesaki theory associates a unique modular operator to each (algebra, cyclic separating vector) pair, and a unitary that intertwines the pairs intertwines the modular operators (Bratteli–Robinson II, §2.5; Takesaki II, Ch. VIII). □

Remark 2. The condition $\eta_i > 0$ enters only through the requirement that a be smooth and strictly positive on $\overline{D_{\mathcal{O}}}$. It cannot be relaxed: as $\eta_i \to 0^+$, a^{-1} blows up and the inverse rescaling $h \mapsto a^{-3} \cdot h$ ceases to be a bijection on C_c^∞ (test functions supported arbitrarily close to $\eta = 0$ are no longer mapped to C_c^∞). The Big Bang singularity is therefore explicitly excluded from the diamond, and the type-II_∞ classification we obtain says nothing about diamonds touching $\eta = 0$. See Question 13 below.

Remark 3. We do *not* consider Rindler-type wedge regions $W = \{(\eta, \mathbf{x}) : x^1 > |\eta - \eta_*|\}$ in radiation-dominated FRW. Any such wedge extends to $\eta \to -\infty$ in Minkowski conformal coordinates; in the FRW chart on $\eta \in (0, \infty)$ this means the wedge unavoidably crosses the singular boundary at $\eta = 0$. Proposition 1 then fails because the wedge has no compact closure on which a is bounded away from 0. For this reason we restrict throughout to causal diamonds bounded away from $\eta = 0$. The same restriction applies to the matter-dominated case (Theorem 6); for slow-roll inflation modelled by exact de Sitter (Theorem 7), the analogous obstruction is that any wedge extends to $\eta \to 0^-$ where $a \to \infty$, which also breaks bijectivity of the rescaling.

3.3 Connes spectrum and the type-II_∞ classification

The Connes spectrum is a W*-invariant of the dynamical system $(\mathcal{M}, \sigma)$ that does not depend on the Hilbert-space realisation. For us the relevant statement is:

Lemma 4. *Let $(\mathcal{M}, \sigma)$ and $(\mathcal{M}', \sigma')$ be W*-dynamical systems and $\alpha : \mathcal{M} \to \mathcal{M}'$ a W*-isomorphism with $\sigma'_t = \alpha \circ \sigma_t \circ \alpha^{-1}$ for all $t \in \mathbb{R}$. Then $\Gamma(\sigma) = \Gamma(\sigma')$.*

Proof. This is immediate from the definition of $\Gamma(\sigma)$ as the intersection over non-zero σ-invariant projections $e \in \mathcal{M}^\sigma$ of the Arveson spectrum of σ restricted to $e\mathcal{M}e$. Both ingredients (the lattice of σ-invariant projections, and the Arveson spectrum on each corner) transfer through α. □

Theorem 5 (Radiation-dominated FRW)**.** *Let φ be the conformally coupled massless scalar field on radiation-dominated FRW with $a(\eta) = a_0\eta$ on $\eta \in (0, \infty)$, in the conformal vacuum Ω_{FRW}. Let $D_{\mathcal{O}}$ be a doubly-bounded causal diamond for a comoving observer with $0 < \eta_i < \eta_f < \infty$, and $\mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}}$ the local von Neumann algebra in the GNS representation. Let σ^{FRW} denote the modular automorphism group of $(\mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}}, \Omega_{\mathrm{FRW}})$. Then:*

1. *$\sigma_t^{\mathrm{FRW}} = \mathrm{Ad}\, U^{-1} \circ \sigma_t^{\mathrm{HL}} \circ \mathrm{Ad}\, U$, where σ^{HL} is the Hislop–Longo modular automorphism of the conformally identified Minkowski diamond $\mathcal{A}(M_{\mathcal{O}})_{\mathrm{M}}$ in the Minkowski vacuum.*

2. $\Gamma(\sigma^{\mathrm{FRW}}) = \mathbb{R}$.

3. *The crossed product* $\mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}} \rtimes_{\sigma^{\mathrm{FRW}}} \mathbb{R}$ *is a type* II_∞ *factor.*

Proof. By Proposition 1, U is a unitary spatially implementing the W*-isomorphism $\mathrm{Ad}\, U : \mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}} \to \mathcal{A}(M_{\mathcal{O}})_{\mathrm{M}}$ with $U\Omega_{\mathrm{FRW}} = \Omega_{\mathrm{M}}$. Both vacua are cyclic and separating for the respective local algebras (standard for free-field vacua on causally complete regions, by the Reeh–Schlieder theorem). Tomita–Takesaki uniqueness then gives $\Delta_{\mathrm{FRW}} = U^{-1}\Delta_{\mathrm{HL}}U$, hence

$$\sigma_t^{\mathrm{FRW}}(A) \;=\; U^{-1}\,\sigma_t^{\mathrm{HL}}\big(UAU^{-1}\big)\,U \qquad \text{for all } A \in \mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}},$$

which is (1).

Applying Lemma 4 with $\alpha = \mathrm{Ad}\, U^{-1}$, $\Gamma(\sigma^{\mathrm{FRW}}) = \Gamma(\sigma^{\mathrm{HL}})$. By Hislop–Longo [7], the Minkowski diamond algebra in the vacuum is a type III_1 factor whose modular automorphism group is the Möbius flow with continuous unbounded modular Hamiltonian K_{HL} (a conformal-Killing-field generator inside the conformal group $\mathrm{SO}(2,4)$ of $\mathbb{R}^{1,3}$, conjugate to a boost generator and hence having spectrum $\mathbb{R}$). Thus $\Gamma(\sigma^{\mathrm{HL}}) = \mathbb{R}$, proving (2).

By Connes–Takesaki, a type III_1 factor crossed by its modular automorphism group with full Connes spectrum $\Gamma = \mathbb{R}$ is type II_∞ [3, 10, §2]. (The condition $\Gamma = \mathbb{R}$ here corresponds to the modular Hamiltonian being unbounded both ways, in line with the $\mathrm{II}_\infty/\mathrm{II}_1$ dichotomy summarised in Section 2.4.) Hence $\mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}} \rtimes_{\sigma^{\mathrm{FRW}}} \mathbb{R}$ is type II_∞, proving (3). □

Theorem 6 (Matter-dominated FRW)**.** *Let* φ *be the conformally coupled massless scalar field on matter-dominated FRW with scale factor* $a(\eta) = a_0\,\eta^2$ *on* $\eta \in (0,\infty)$*, in the conformal vacuum* Ω_{FRW}*. Let* $D_{\mathcal{O}}$ *be a doubly-bounded causal diamond for a comoving observer with* $0 < \eta_i < \eta_f < \infty$*, and* $\mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}}$ *the local von Neumann algebra in the GNS representation. Then conclusions (1)–(3) of Theorem 5 hold verbatim with the same proof.*

Proof. The metric is again conformally flat in the form (1), the scalar field equation is conformally invariant by (9), and on the closed diamond $\overline{D_{\mathcal{O}}}$ the scale factor $a(\eta) = a_0\eta^2$ is smooth and strictly positive (since $\eta_i > 0$, a is bounded away from 0 on $\overline{D_{\mathcal{O}}}$, and a is bounded above as well since $\eta_f < \infty$). The constructions of Section 3.1 and Proposition 1 therefore go through unchanged. The conclusions follow exactly as in Theorem 5. □

Theorem 7 (Slow-roll inflation modelled by exact de Sitter)**.** *Let* φ *be the conformally coupled massless scalar field on the de Sitter spacetime in flat-slicing conformal coordinates, with scale factor* $a(\eta) = -1/(H\eta)$ *on* $\eta \in (-\infty, 0)$*,* $H > 0$*, in the conformal vacuum* Ω_{FRW}*. Let* $D_{\mathcal{O}}$ *be a doubly-bounded causal diamond for a comoving observer with* $-\infty < \eta_i < \eta_f < 0$*, and* $\mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}}$ *the local von Neumann algebra in the GNS representation. Then conclusions (1)–(3) of Theorem 5 hold verbatim with the same proof.*

Proof. On the closed diamond $\overline{D_{\mathcal{O}}}$ with $\eta_i, \eta_f \in (-\infty, 0)$, the scale factor $a(\eta) = -1/(H\eta)$ is smooth and strictly positive: indeed $\eta < 0$ on $\overline{D_{\mathcal{O}}}$, so $-H\eta > 0$ and $a = 1/(-H\eta) > 0$; smoothness fails only at $\eta = 0$, which is excluded by $\eta_f < 0$. Moreover, a is bounded above on $\overline{D_{\mathcal{O}}}$ (by $1/(-H\eta_f) < \infty$) and bounded below (by $1/(-H\eta_i) > 0$). The metric $\mathrm{d}s^2 = a(\eta)^2(-\mathrm{d}\eta^2 + \mathrm{d}\mathbf{x}^2)$ is conformally flat by inspection. The conformal-covariance identity (9) holds for any smooth a, in particular for $a(\eta) = -1/(H\eta)$ on $\eta < 0$ (this is the standard Poincaré patch of de Sitter; the Ricci scalar evaluates to $R = 12H^2 > 0$, but the identity (9) is off-shell and depends only on the conformal-flatness structure, not on R being constant). The constructions of Section 3.1 and Proposition 1 therefore go through unchanged. The conformally coupled massless scalar on de Sitter

has a well-defined Bunch–Davies / conformal vacuum [9, §5.4], which by definition coincides with the conformal pullback of the Minkowski vacuum on $M_{\mathcal{O}}$. The conclusions follow exactly as in Theorem 5. □

Remark 8. Theorem 7 models the inflationary epoch by exact de Sitter; for slow-roll inflation one has $a(\eta) \approx -1/(H\eta)$ to leading order in the slow-roll parameters, with corrections that break exact conformal flatness of the FRW background only via the time-dependence of H. To the extent that an FRW background remains conformally flat (which it does for any spatially flat FLRW), the symplectic and covariance identities of Section 3.1 continue to hold and the theorem still applies; what we strictly need from $a(\eta)$ is smoothness and strict positivity on $\overline{D_{\mathcal{O}}}$, both of which are satisfied for any FRW background with no horizon crossing inside the closed diamond. Boundedness of $a(\eta)$ on $\overline{D_{\mathcal{O}}}$ is not required by the proof, only positivity (and smoothness): the rescaling $f \mapsto a^3 f$ is a C_c^∞-bijection as soon as a and a^{-1} are both C^∞ on $\overline{D_{\mathcal{O}}}$.

Remark 9. None of the FRW spacetimes considered above admits a timelike Killing vector field along a generic comoving worldline (radiation-dominated and matter-dominated FRW have no timelike Killing field at all; de Sitter has timelike Killing fields only inside static patches, not along comoving worldlines of the flat slicing). The proofs of Theorems 5, 6, 7 do not invoke any timelike Killing field of g_{FRW}; the modular flow is identified by a W*-algebraic argument that uses only the conformal isometry to Minkowski plus the Hislop–Longo theorem in the Minkowski vacuum.

4 Comparison with CLPW and Witten

The following table summarises the structural differences (all entries refer to causally complete regions: wedges in the Killing-horizon settings, doubly-bounded diamonds in the present cosmological setting):

Setting	Modular flow σ implements	Crossed product type
Minkowski wedge [1, 2]	boost (Killing)	III_1 (no crossed product)
Minkowski diamond [7]	Möbius (conformal Killing)	III_1 (no crossed product)
de Sitter static patch [3]	dS boost (Killing)	II_1
Schwarzschild exterior [10]	Schwarzschild time (Killing)	II_∞
General Killing horizon [5]	Killing generator	II_∞
Radiation/matter-dominated FRW (this note)	**conformal pullback of HL**	$\mathbf{II}_\infty$
Slow-roll inflation (this note)	**conformal pullback of HL**	$\mathbf{II}_\infty$

In CLPW [3], the de Sitter result uses the dS boost on the static patch as a Killing vector field; the modular flow is implemented by this Killing field, and the bounded-below modular Hamiltonian forces the crossed product to be type II_1 rather than II_∞. In Witten's earlier paper [10], the Schwarzschild result uses the timelike Killing field outside the horizon; the unbounded modular Hamiltonian on the exterior gives type II_∞. Faulkner–Speranza [5] extend this to general Killing horizons with a non-perturbative GSL.

In all three of these settings the modular flow is geometrically implemented by a Killing vector field of the background, which is essential for the explicit identification of σ as a flow. The cosmological cases of the present note are structurally different: there is no timelike Killing vector along the comoving worldline (radiation- and matter-dominated FRW have none at all; de Sitter has them only

inside the static patch, not along the comoving worldlines used here), and the modular flow is *not* the flow of any vector field of g_{FRW}. It is, however, the conjugate (by the conformal-rescaling unitary U of Proposition 1) of the Hislop–Longo Möbius flow on the Minkowski diamond, which *is* a flow of a conformal Killing field of the Minkowski metric $\eta_{\mu\nu}$ obtained from g_{FRW} by conformal rescaling. The classification result then follows from Tomita–Takesaki uniqueness (the unitary U intertwines the modular operators) and W*-isomorphism invariance of the Connes spectrum (Lemma 4), rather than from an identification of σ as a geometric flow on the FRW spacetime.

This is, to our knowledge, the first explicit example of a CLPW-type II_∞ classification for cosmological backgrounds that are not stationary in any geometric sense. It is consistent with [5] where Killing structure is essential, in the trivial sense that the present setting is outside the scope of that work; and consistent with [3, 10] as a different proof strategy for an analogous classification.

Priority check (2026-05-02)

To verify the priority claim of the previous paragraph we conducted a literature search via the arXiv API, run independently of the manuscript, with the following queries (all restricted to abstract/full-text indexing): `"crossed product" AND "conformal" AND ("FRW" OR "cosmological")`; `"type II" AND "von Neumann" AND "cosmology"`; `"crossed product" AND "FLRW"`; `"conformally coupled" AND "crossed product"`; `"Bunch-Davies" AND "crossed product"`; `"Hislop-Longo" AND ("cosmology" OR "FRW" OR "FLRW" OR "de Sitter")`; `"modular flow" AND "FRW"`; `"FRW" AND ("Tomita" OR "modular operator")`; `"Bisognano-Wichmann" AND ("cosmology" OR "FLRW" OR "expanding")`; `"radiation dominated" AND "type II" AND "algebra"`; `"conformally flat" AND "modular" AND "crossed product"`; `"non-stationary" AND "von Neumann" AND "modular"`; `"conformal isometry" AND "FRW" AND "quantum field"`; `"conformal pullback" AND "modular"`. Six of the most directly worded queries (those simultaneously combining *crossed product* with *FRW/FLRW* or *conformally coupled* or *Bunch–Davies*, and the dedicated *Hislop–Longo*+cosmology query) returned zero hits; `"FRW" AND ("Tomita" OR "modular operator")` also returned zero hits. The other queries returned the post-CLPW corpus, which we then read at the abstract level.

The closest related works we identified are the following:

- *Cosmology as a Crossed Product*, C. Gomez, arXiv:2207.06704; *Entanglement, Observers and Cosmology*, C. Gomez, arXiv:2302.14747; *Clocks, Algebras and Cosmology*, C. Gomez, arXiv:2304.11845. Treats inflationary cosmology by adding a clock to the de Sitter static-patch type III_1 algebra; the underlying spacetime is de Sitter and the modular flow is the de Sitter Killing boost. No conformal pullback to Minkowski Bisognano–Wichmann; no FRW.

- *Von Neumann algebra description of inflationary cosmology*, S. Kawamoto, T. Mori, S. Takahashi, arXiv:2212.05637 (and the 2026 follow-up arXiv:2603.25990 by similar authors). Establishes type II_∞ for inflationary quasi-de Sitter via a Stückelberg / spontaneously-broken-isometry mechanism, in the static-patch description. The classification is for quasi-de Sitter, not radiation-dominated FRW; the technique is fundamentally different (Stückelberg dressing, not conformal pullback) and the conformal-coupling assumption is not used.

- *A clock is just a way to tell the time: gravitational algebras in cosmological spacetimes*, C.-H. Chen, G. Penington, arXiv:2406.02116. Constructs a type II_∞ algebra for slow-roll inflation and evaporating Schwarzschild–de Sitter using the rolling inflaton (or evaporating black hole) as a physical clock; explicitly notes that the resulting algebras "are both Type II_∞ factors, but neither is manifestly a crossed product algebra." No use of conformal flatness or conformal pullback; radiation-dominated FRW is not treated.

- *An intrinsic cosmological observer*, arXiv:2504.07630. Constructs an observer intrinsically from quantum fields in the slow-roll setup of [4] and connects to Connes–Takesaki flow of weights. Applies to slow-roll inflation, not to conformally flat radiation-dominated FRW.

- *Modular Hamiltonian for de Sitter diamonds*, M. B. Fröb, arXiv:2308.14797 (JHEP 12 (2023) 074). Computes the Tomita–Takesaki modular Hamiltonian for CFTs in double-cone and light-cone regions of conformally flat spacetimes, with the de Sitter diamond as the principal worked example. This is structurally the closest precedent to the conformal-pullback step used in our Proposition 1, but Fröb's paper (i) uses CFT axioms rather than the conformally coupled massless scalar specifically, (ii) works in de Sitter, not radiation-dominated FRW, (iii) does not classify the resulting algebra as type III_1 or construct a crossed product, and (iv) makes no contact with CLPW or with the type II_∞ statement. The companion paper *Relative Entropy in de Sitter is a Noether Charge*, M. B. Fröb, arXiv:2310.12185, similarly restricts the Minkowski wedge modular operator to a de Sitter hyperboloid, but again does not address either FRW or the crossed-product classification.

- *Gravitational algebras and the generalized second law*, T. Faulkner, A. J. Speranza, arXiv:2405.00847; *Generalised second law beyond the semiclassical regime*, arXiv:2412.01903. Both works explicitly require "a state whose modular flow is geometric on the horizon," i.e. a Killing horizon; radiation-dominated FRW with a comoving observer admits no such Killing structure, so these results do not cover the present setting.

- Earlier algebraic-QFT works on FRW by Dappiaggi–Moretti–Pinamonti (e.g. arXiv:0712.1770, 0801.2850, 0812.4033, 1009.5179) construct Hadamard / KMS-type states for free quantum fields in spatially flat FRW. They do not address the type-classification of local algebras and predate CLPW; they make no crossed-product construction.

Verdict. The conformal-pullback identification of the FRW algebra of a comoving diamond with the Minkowski Bisognano–Wichmann / Hislop–Longo algebra in the conformally coupled massless scalar case is a strict consequence of (i) classical conformal flatness of FRW in conformal coordinates and (ii) classical conformal invariance of the conformally coupled massless scalar in $d = 4$. As such the technical content of Proposition 1 is folklore in algebraic QFT on curved spacetime. What the present note adds is the explicit *combination* of this folklore with the post-CLPW type-classification machinery (Connes–Takesaki) to obtain a type II_∞ classification for the gravitationally dressed algebra of a non-stationary cosmological wedge / diamond. Within the literature surveyed above we have not been able to locate a statement of Theorem 5 (or its parts (1)–(3)) in this form. The closest precedent (Fröb 2023) computes modular data in conformally flat spacetimes for de Sitter only and stops short of the type-classification step; the closest type-II_∞ cosmological results (Gomez; Kawamoto et al.; Chen–Penington; intrinsic observer 2504.07630) all work with quasi-de Sitter / inflation and use a clock or Stückelberg dressing rather than a conformal pullback; the FRW Hadamard-state literature (Dappiaggi–Pinamonti and collaborators) predates CLPW and does not address types. We therefore retain the priority claim of the preceding paragraph in the conservative form: *first explicit recording, with explicit proof and explicit Connes-spectrum invariance step, of the conformal-pullback type II_∞ classification for the conformally coupled massless scalar on radiation-dominated (and matter-dominated) FRW.* We make no claim of independence from [6] in the modular-data step, nor of priority over the quasi-de Sitter type II_∞ classifications of [4, 8] which use disjoint techniques.

5 Open questions

The argument above is sharp on the conformal trick. Several natural questions remain.

Open Question 10 (Massive scalar)**.** For $m > 0$, the field equation $(\Box_g - R/6 - m^2)\varphi = 0$ does *not* reduce to $\Box_{\tilde{g}}\,\tilde{\varphi} = 0$ under the conformal rescaling: an effective mass term $m^2 a(\eta)^2$ remains in the rescaled equation. The unitary U does not intertwine the field dynamics, the FRW vacuum is not the conformal pullback of the Minkowski vacuum, and Theorem 5 does not apply. The type of the crossed product for massive scalars in radiation-dominated FRW is open.

Open Question 11 (Non-conformal scalars and interacting fields)**.** The argument relies on the conformal coupling $\xi = 1/6$ in $d = 4$. For $\xi \neq 1/6$ the rescaling (4) fails to map the FRW field equation to the flat-space wave equation, and Proposition 1 fails. Interacting fields ($\lambda\varphi^4$, gauge couplings) also break conformal invariance and are completely outside the scope of the present argument.

Open Question 12 (Non-conformally-flat backgrounds)**.** The argument requires conformal flatness of the background metric. Spatially flat FRW spacetimes (radiation, matter, or mixed conformal-matter content with vanishing Weyl tensor) are conformally flat in conformal coordinates. Spatially open or closed FRW slicings (and Bianchi cosmologies more generally) are typically *not* conformally flat: the Weyl tensor is non-zero. For such backgrounds the conformal trick does not apply, and a different strategy is required.

Open Question 13 (Big Bang)**.** The unitary U of Proposition 1 requires a to be smooth and strictly positive on $\overline{D_{\mathcal{O}}}$, which forces $\eta_i > 0$ in radiation- and matter-dominated FRW (and $\eta_f < 0$ for de Sitter). For diamonds touching the Big Bang singularity ($\eta_i = 0$) or the conformal boundary in de Sitter ($\eta_f = 0$), the rescaling map $f \mapsto a^3 \cdot f$ is no longer a C_c^∞-bijection, and the present argument fails. The classification of $\mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}} \rtimes_\sigma \mathbb{R}$ for such diamonds is open.

Open Question 14 (Backreaction)**.** The argument is at the level of QFT in a fixed classical FRW background. The crossed-product algebra $\mathcal{A}(D_{\mathcal{O}}) \rtimes_\sigma \mathbb{R}$ models a single quantum clock degree of freedom dressed against the matter algebra; backreaction of this clock plus the matter content on the FRW geometry is not addressed. The CLPW interpretation [3] of the type-II structure as the algebra of dressed gravitational observables in a semiclassical regime should carry over without modification, but a fully gravitational treatment (allowing the metric to fluctuate) is not part of the present argument.

Open Question 15 (Killing-free cosmological observable algebras)**.** Theorem 5 is a single-case-study result. A natural research programme is to formulate and prove a general criterion: *for which classes of (possibly non-stationary) cosmological backgrounds does the gravitational crossed-product algebra of a comoving observer fall under a CLPW-type classification?* The conformal-flatness hypothesis used here is sufficient but possibly not necessary. A satisfying answer would be a structural theorem expressed in terms of invariants of the local algebra (e.g. the Arveson spectrum of the modular automorphism, viewed as a function of the geometry of the diamond), not the existence of a Killing vector field.

Acknowledgements

This note records a result first observed in commit `aa9a3ee` of the `crossed-cosmos` repository (2026-05-02), during a multi-hour audit session. The author thanks the anonymous mathematical-physics community on arXiv whose works [1–3, 5, 7, 10] are the foundation on which this short observation builds.

References

[1] Joseph J. Bisognano and Eyvind H. Wichmann. On the duality condition for a Hermitian scalar field. *J. Math. Phys.*, 16(4):985–1007, 1975.

[2] Joseph J. Bisognano and Eyvind H. Wichmann. On the duality condition for quantum fields. *J. Math. Phys.*, 17(3):303–321, 1976.

[3] Venkatesa Chandrasekaran, Roberto Longo, Geoff Penington, and Edward Witten. An algebra of observables for de sitter space. *JHEP*, 02:082, 2023.

[4] Chang-Han Chen and Geoff Penington. A clock is just a way to tell the time: gravitational algebras in cosmological spacetimes. 2024.

[5] Thomas Faulkner and Antony J. Speranza. Gravitational algebras and the generalized second law. *JHEP*, 11:099, 2024.

[6] Markus B. Fröb. Modular Hamiltonian for de Sitter diamonds. *JHEP*, 12:074, 2023.

[7] Peter D. Hislop and Roberto Longo. Modular structure of the local algebras associated with the free massless scalar field theory. *Commun. Math. Phys.*, 84(1):71–85, 1982.

[8] Shoichi Kawamoto, Tsunehide Mori, and Sotaro Takahashi. Von Neumann algebra description of inflationary cosmology. 2022.

[9] Robert M. Wald. *Quantum Field Theory in Curved Spacetime and Black Hole Thermodynamics*. Chicago Lectures in Physics. University of Chicago Press, Chicago, IL, 1994.

[10] Edward Witten. Gravity and the crossed product. *JHEP*, 10:008, 2022.